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In-domain adaptation of FinBERT to retail-investor social-media text: A controlled test of explainability-guided data selection

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Abstract: Domain-adaptive transformer models such as FinBERT are widely used for financial sentiment classification, but they are typically trained and evaluated on curated, professionally written text. We examine how FinBERT performs on the noisier text that a deployed system ingests, retail-investor discussion on Reddit, posts on X/Twitter and news, and whether a small amount of in-domain fine-tuning helps. On a 338-item benchmark sampled from the system's own multi-source database, with model-assisted labels audited by a human annotator, stock FinBERT reaches a macro-F1 of 0.498 and over-predicts the neutral class. We fine-tuned FinBERT on three 900-item training arms, each over five seeds. Two arms share a 600-item positive/negative base and differ only in their neutral examples: one uses neutral text containing sentiment-loaded vocabulary, a selection motivated by local interpretable model-agnostic explanations (LIME), and the other uses randomly chosen neutral text as a control. Mean macro-F1 rose to 0.591 for the LIME-guided arm and 0.607 for the random-neutral arm, gains of 9.3 and 11.0 points over stock FinBERT that hold under an item-level paired bootstrap with Holm correction (adjusted $p = 0.004$ and $p < 0.001$). A third, mixed arm (0.534) gave no reliable gain. LIME-guided selection did not outperform the random control (difference -1.6 points, 95% CI $[-5.0, +1.7]$). Out of domain, every fine-tuned model scored below stock FinBERT on Financial PhraseBank and on Twitter Financial News (for example, macro-F1 0.630 against 0.668 on the latter for the random-neutral arm). The results support modest in-domain adaptation, not improved general-purpose classification, and show the value of testing explanation-derived hypotheses against a matched control.

Keywords: financial sentiment analysis; FinBERT; explainable AI; LIME; domain adaptation; fine-tuning; social media; investor sentiment; multi-source text; model error analysis
JEL Codes: C45, C52, C53, G14, G17, G41

1. Introduction

Markets do not move on facts alone; they move on how investors read those facts. Investor sentiment is a well-documented driver of asset prices, trading volume and volatility, capable of producing temporary mispricings that persist until information is absorbed (Tetlock, 2007; Baker and Wurgler, 2006). The migration of investor discourse onto social media and online communities has made it possible to quantify this sentiment at scale, and domain-adaptive transformer models, chiefly FinBERT (Araci, 2019), have become a standard tool for doing so, capturing implicit financial phrasing ("guidance raised", "missed expectations", "strong headwinds") that earlier lexicon and bag-of-words methods cannot.

FinBERT was, however, fine-tuned on curated financial news sentences, and its behaviour on retail-investor social-media posts, which are shorter, more informal and differently balanced across classes, is less well characterised. Its handling of the *neutral* class is a frequently reported weakness, but the direction and size of neutral-class errors depend on the text distribution and the labelling conventions used. Existing surveys of explainable AI (XAI) in finance also note that interpretability methods are mostly used for post-hoc presentation or offline auditing rather than to inform model improvement (Yeo et al., 2025), and such uses are rarely tested against a control.

This paper asks two questions. First, does a small amount of in-domain fine-tuning improve FinBERT on the multi-source text that a deployed system actually ingests? Second, does choosing fine-tuning data on the basis of local explanations help more than choosing it at random? We make the following contributions:

1. **A multi-source evaluation benchmark.** We construct a 338-item benchmark sampled from a deployed system's database of Reddit, X/Twitter and news text, with model-assisted labels under a written rubric and a human audit. On it, stock FinBERT reaches a macro-F1 of 0.498 and over-predicts the neutral class (228 neutral predictions against 135 gold neutral items).
2. **A controlled in-domain fine-tuning comparison.** We fine-tune FinBERT on three 900-item arms over five seeds each and compare them with stock FinBERT using paired tests with multiple-comparison correction. Mean macro-F1 rises to 0.591 and 0.607 for the two arms that share a positive/negative base (bootstrap gains of 9.3 and 11.0 points, Holm-adjusted $p = 0.004$ and $p < 0.001$), while a mixed arm (0.534) gives no reliable gain.
3. **A test of LIME-guided data selection against a random control.** LIME examples suggested that neutral text containing sentiment-loaded vocabulary would be especially useful for fine-tuning. Selecting neutrals this way did not outperform randomly selected neutrals (difference -1.6 points, 95% CI $[-5.0, +1.7]$), so we treat LIME here as hypothesis-generating rather than as the source of the improvement.
4. **An out-of-domain check.** All fine-tuned models score below stock FinBERT on Financial PhraseBank and Twitter Financial News. We report this as a limitation: the gains are specific to the target distribution and are not a claim of better general financial sentiment classification.

These results were obtained within, and are reported alongside, a fully deployed analytics system. The system's additional components (a finance-emotion enrichment layer, a lightweight aspect extractor, and a sentiment-price correlation engine) are described for contextual completeness, and their evaluations, including negative and null findings, are reported transparently rather than presented as

headline claims. The remainder of the paper is organised as follows. Section 2 reviews related work; Section 3 describes the data; Section 4 sets out the methodology; Section 5 presents results; Section 6 discusses implications and threats to validity; and Section 7 concludes.

2. Literature review

2.1. Sentiment, text and markets

Behavioural finance establishes that sentiment and psychological factors produce temporary, economically meaningful mispricings (Baker and Wurgler, 2006). Tetlock (2007) showed that negative language in financial news predicts short-term downward pressure on returns with subsequent reversion; Antweiler and Frank (2004) had already established that user-generated message-board text carries statistically significant information about volatility, and Schumaker and Chen (2009) reported comparable short-horizon predictive content in breaking financial news. Bollen et al. (2011) extended this to social media at scale, deriving daily mood indices from Twitter that predicted Dow Jones movements, while Da et al. (2015) reached a similar conclusion from aggregate search volumes, and Garcia (2013) showed news-sentiment effects are strongest during recessions. More recent work documents the distinct behaviour of retail communities: Long et al. (2023) show that Reddit sentiment during the GameStop episode diverged sharply from institutional news, and cryptocurrency markets exhibit comparable retail-driven sensitivity to Twitter sentiment (Kraaijeveld and De Smedt, 2020). The collective implication is that single-source sentiment signals are structurally incomplete, motivating multi-source measurement.

2.2. From lexicons to domain-adaptive transformers

Early financial sentiment systems relied on domain-specific lexicons; Loughran and McDonald (2011) demonstrated that general-purpose word lists misclassify finance vocabulary and produced the finance-specific dictionary that remains the standard lexicon baseline. Lexicon methods are transparent and fast but fail on negation, implicit polarity and context-dependent phrasing. The transformer architecture (Vaswani et al., 2017) and bidirectional pre-training (Devlin et al., 2019) changed the field, and FinBERT (Araci, 2019) adapted this to finance by pre-training on financial corpora, consistently outperforming general models and lexicons on the Financial PhraseBank benchmark (Malo et al., 2014). Parallel FinBERT variants target financial communications and filings (Yang et al., 2020; Huang et al., 2023), and very recent work continues to adapt these models, for example, through pseudo-label test-time adaptation of FinBERT (Yang et al., 2026) and sparse mixture-of-experts augmentation of LLM-based sentiment classifiers (Li et al., 2026).

Since 2023, LLMs have been applied to financial sentiment with mixed but informative results. ChatGPT-based scores have been found to outperform FinBERT on some benchmarks and to carry incremental return-predictive information (Fatouros et al., 2023; Lopez-Lira and Tang, 2023), while other work shows LLM sentiment is brittle under adversarial rephrasing (Leippold, 2023) and unreliable on informal Reddit-register text (Deng et al., 2023). Domain LLMs such as BloombergGPT (Wu et al., 2023) demonstrate strong zero-shot performance but are not publicly available, and generative models are increasingly trialled across financial decision workflows beyond sentiment, for instance, in venture screening (Vismara et al., 2026). The consensus is that LLMs have narrowed the per-sentence

accuracy gap with FinBERT on clean benchmarks but have not displaced it where latency, cost and reproducibility are first-order constraints, which is the setting of this study. Across this literature, the neutral class is repeatedly identified as the hardest and least-studied category, but is rarely diagnosed or remediated directly.

2.3. Explainable AI in finance: presentation versus diagnosis

Doshi-Velez and Kim (2017) frame interpretability as explaining model behaviour in human-understandable terms wherever objectives cannot be fully specified in advance, which is almost always the case in finance. LIME (Ribeiro et al., 2016) explains individual predictions by fitting a sparse linear surrogate over perturbations of the input, identifying the tokens most responsible for a prediction; Shapley additive explanations (SHAP) (Lundberg and Lee, 2017) provides game-theoretic guarantees at higher computational cost. Surveys find that XAI in finance is dominated by post-hoc LIME and SHAP but that deployment within interactive tools, and use as a feedback mechanism, remain rare (Yeo et al., 2025). Sravankumar et al. (2025) integrate XAI with deep learning for sentiment-driven stock prediction. Two cautionary results bound the claims that interpretability can support: perturbation-based explanations can be adversarially fooled (Slack et al., 2020), and intrinsically interpretable models may be preferable in high-stakes settings (Rudin, 2019). We therefore use LIME strictly as a *diagnostic aid*, a hypothesis-generating instrument whose output is corroborated against the confusion matrix and an intervention, rather than as a guarantee of correctness. The gap this paper addresses is thus narrow and specific: the use of local interpretability to *diagnose a known model failure mode precisely enough to drive a targeted, validated mitigation*, evaluated on realistic multi-source financial text.

3. Data

3.1. Multi-source corpus

The deployed system ingests text from three complementary channels via documented official interfaces: Reddit (long-form discussion from finance subreddits, via PRAW), X/Twitter (high-frequency reactions, via the official X API v2), and financial news (professionally sourced articles, via NewsAPI). All collection is restricted to publicly available text; author-identifying fields (handles, display names, geolocation) are stripped at ingestion, so that each stored record contains only sanitised text, a timestamp, a source identifier, and any resolved ticker mentions. Duplicate and near-duplicate detection (content-hashing and TF-IDF cosine similarity) and minimum-length thresholds are applied. At the time of analysis, the corpus comprised more than 150,000 labelled records. We note a material data-layer limitation up front: tightened X API v2 quotas during 2023 to 2024 bounded the Twitter subcorpus well below the Reddit and news subcorpora, so the Twitter signal is treated as secondary and corroborated throughout (see Section 6). The project received a favourable ethical opinion from the Loughborough University Ethics Review Sub-Committee through the University's ethical review system; full approval and consent details are given in the Ethics statement.

3.2. Evaluation benchmark and label validation

Existing benchmarks such as Financial PhraseBank (Malo et al., 2014) consist of professionally written news sentences and do not reflect the informal, noisy language the deployed system processes. We therefore constructed a benchmark (benchmark v2) directly from the system's database of ticker-tagged records (159,081 records at the time of sampling). Records were cleaned (URLs removed, whitespace normalised), restricted to sentence-scale texts of 60 to 500 characters with a heuristic English filter, and de-duplicated on normalised text. An initial sample of 300 items was drawn uniformly at random within each channel (100 Reddit, 100 news, 100 Twitter; fixed seed), without reference to any model output. Items labelled out of scope in this draw were replaced by further random draws from the same channel (16 drawn; 12 retained), and, because the random draw contained few negative items, a top-up of negative-lexicon-matching items was added (60 drawn; 56 retained); this top-up is the only non-random component of the benchmark and is recorded per item. After removal of 21 items judged unjudgeable or out of scope, the frozen benchmark contains 338 items (135 neutral, 122 positive, 81 negative; 108 Reddit, 115 news, 115 Twitter), of which 66 are flagged as low-information factual filler and retained as neutral under the labelling rubric. Post timestamps (the stored publication time, present for all 338 items) span 24 August 2017 to 13 August 2026: news items from 24 August 2017 to 15 July 2026, Reddit items from 8 February 2021 to 13 August 2026, and Twitter items from 3 January 2022 to 6 April 2026. The class mix and ticker coverage per channel are reported in Table 6.

Labels were assigned by model-assisted labelling under a written rubric (rubric v2), with human audit by one author (D.H.). The rubric defines sentiment as the investor view of the asset discussed and fixes boundary cases (price-move and chart calls follow the stated direction; mixed price-target lists and factual filler are neutral). On a held-out reserve audit of 31 classifiable items, not used for rubric development, agreement between D.H. and the model-assisted labels was Cohen's $\kappa = 0.807$ (Cohen, 1960) (87.1% agreement). The original 200-item benchmark used in an earlier version of this work was not sampled from the corpus and is no longer used for any reported result.

3.3. Fine-tuning corpus

The 300-item fine-tuning corpus used in an earlier version of this study, and the run behind its reported result, could not be recovered; all fine-tuning results reported here are new. A candidate pool of 2,300 posts was drawn from the same database with the same cleaning and length filters, excluding every benchmark item by identifier and normalised text, in two fixed-seed draws (1,100 then 1,200 items; in each, roughly 36% unrestricted random and 64% matching at least one term of the system's finance sentiment lexicon). All 2,300 candidates were labelled with rubric v2 by the same model-assisted procedure: 979 positive, 593 neutral, 635 negative and 93 excluded as out of scope. To audit the label most relevant to this study, D.H. blind-relabelled a random sample of 30 candidates labelled neutral: agreement was 23/30, and neutral precision was 23/29 (0.793) after removing one item D.H. judged out of scope (the remaining disagreements were 5 positive and 1 negative).

Three training arms were built from this pool (Table 1). All three share an identical base of 300 positive and 300 negative items, sampled with a fixed seed from labelled positives and negatives not used elsewhere in the arms. They differ only in a 300-item arm-specific component: (a) *lexicon-loaded neutrals*, neutral items containing at least one term of the finance sentiment lexicon, the vocabulary that the LIME analysis (Section 4.4) implicates in neutral errors; (b) *random neutrals*, drawn at random

from all labelled neutrals and disjoint from arm (a) where possible (7 items shared), as a control for neutral exposure without lexical targeting; and (c) a *mixed* component of 100 lexicon-loaded neutrals, 100 positives and 100 negatives, which replays polarised examples instead of adding neutral mass. The shared base is a deviation from the pre-registered plan, in which each arm consisted of its 300-item component alone: arms (a) and (b) would then contain a single class, and a one-epoch pilot on arm (a) alone predicted *neutral* for every benchmark item. A consequence of the base is that arm (c) differs from (a) and (b) in class proportions (400/400/100 rather than 300/300/300), which is the intended class-prior contrast but should be borne in mind when comparing it with the other arms. Every arm is disjoint from the benchmark by identifier and normalised text, and all arms are rebuilt byte-identically from the build script.

Table 1. Composition of the three fine-tuning arms (900 items each).

Arm	Arm-specific component	Positive	Negative	Neutral	Low-info.
(a) Lexicon-loaded	300 lexicon-loaded neutrals	300	300	300	87
(b) Random-neutral	300 random neutrals	300	300	300	105
(c) Mixed	100 loaded neu. + 100 pos. + 100 neg.	400	400	100	31

4. Methodology

4.1. Classifier and baseline

The primary classifier is FinBERT, loaded from the public `ProsusAI/finbert` checkpoint and used in zero-shot inference (no task-specific fine-tuning in the main comparison). It outputs a three-class label (positive/neutral/negative) with a softmax probability distribution, which is retained in full for the downstream enrichment described in Section 4.6. The baseline is a finance-lexicon keyword classifier built on domain-specific vocabulary, representing the best-practice lexicon approach cf. (Loughran and McDonald, 2011) and providing both a meaningful lower anchor and a fast fallback. Text from all sources passes through a shared preprocessing pipeline tuned for transformer input: case preservation, URL/markup removal, stopword retention, no lemmatisation, preservation of financial punctuation, and negation rewriting.

4.2. Evaluation metrics

We report accuracy; macro precision, recall and F1 (equal weight per class, which matters here because the benchmark is not class-balanced); per-class precision, recall and F1; predicted-class counts; and confusion matrices. For a class c with precision P_c and recall R_c, the macro-F1 is given by:

$$\text{macro-F1} = \frac{1}{|C|} \sum_{c \in C} \frac{2P_c R_c}{P_c + R_c},$$

where $C = \{\text{positive}, \text{negative}, \text{neutral}\}$ is the set of sentiment classes and $|C| = 3$.

4.3. Paired significance testing

All models are evaluated on the same 338 benchmark items, so comparisons are paired. We use three complementary procedures. (i) For each fine-tuned run against stock FinBERT, McNemar's test on discordant items (Dietterich, 1998), using the exact two-sided binomial p-value; with b items correct only for one model and c only for the other, the continuity-corrected statistic is

$$\chi^2_{cc} = \frac{(|b - c| - 1)^2}{b + c}.$$

(ii) For each arm against stock and each pair of arms, an item-level paired bootstrap (Efron and Tibshirani, 1993) of the difference in macro-F1, where each arm's macro-F1 on a resample is the mean over its five seeds ($B = 10{,}000$ resamples of the 338 items, fixed seed; two-sided p from the bootstrap distribution of the difference). (iii) For arm-versus-arm comparisons, seed-paired t-tests and exact Wilcoxon signed-rank tests over the five seeds. With five seeds the exact two-sided Wilcoxon test cannot reach $p < 0.0625$, so these tests have little power and are reported as secondary. Holm's step-down correction is applied within each family of tests (15 McNemar tests; six bootstrap comparisons; three seed-paired comparisons per metric and test).

4.4. LIME as a diagnostic instrument

To explain individual predictions we apply LIME (Ribeiro et al., 2016), which perturbs the input by masking tokens, queries FinBERT on the perturbed variants, and fits a sparse local linear surrogate whose weights identify the tokens most responsible for the prediction. We use LIME to generate a hypothesis about *why* neutral classification fails, by inspecting which tokens drive the predicted polarity on representative neutral texts. The random state is fixed for reproducibility of the reported explanations. Consistent with Slack et al. (2020) and Rudin (2019), LIME output is treated as illustrative and hypothesis-generating rather than as proof; the hypothesis it suggests is tested by the controlled comparison of arms (a) and (b).

4.5. Fine-tuning protocol

Each arm was used to fine-tune the base `ProsusAI/finbert` checkpoint with each of five seeds (13, 21, 42, 84, 168), giving 15 runs. For each run, the arm was split 90/10 into training and validation sets, stratified by class, with the run's seed. Training used AdamW (learning rate 2×10^{-5}, weight decay 0.01, not applied to bias and layer-normalisation parameters), a linear schedule with warm-up over the first 10% of steps, batch size 16, gradient-norm clipping at 1.0, and a maximum sequence length of 128 tokens, for three epochs. The epoch with the highest validation macro-F1 was selected (ties to the earlier epoch), and only that epoch's model was evaluated on the benchmark; the benchmark played no part in model selection. Stock FinBERT was evaluated on the benchmark with the same 128-token limit (a 512-token limit changed macro-F1 by less than 0.3 points). Training ran on a laptop CPU (Intel Core i9-9980HK) and took about 12 minutes per run, including per-epoch evaluation.

Out-of-domain checks. To test whether in-domain gains transfer, stock FinBERT and one model per arm were also evaluated on two external datasets: Financial PhraseBank (Malo et al., 2014), read from the official release, in its all-agree (2,264 sentences) and 50%-agree (4,846 sentences) subsets; and the

validation split of the Twitter Financial News sentiment dataset (2,388 items; labels mapped Bearish → negative, Bullish → positive, Neutral → neutral), which shares no text with any arm or with the benchmark. For each arm the seed with the highest validation macro-F1 was used (arm (a) seed 84, arm (b) seed 84, arm (c) seed 21); these models were retrained with identical settings and reproduced the original benchmark predictions exactly. Because `ProsusAI/finbert` was itself fine-tuned on Financial PhraseBank, stock FinBERT's PhraseBank scores are in-sample and therefore inflated.

4.6. Supporting system components

For completeness we summarise three components that constitute the deployed system but are not the paper's primary contribution. (i) A *finance-emotion enrichment layer* maps FinBERT's softmax vector, a normalised Shannon-entropy term, domain lexicon matches and aspect priors onto six behavioural emotion classes (fear, optimism, uncertainty, confidence, scepticism, mixed) in a single inference pass, without labelled emotion data. The entropy term, which lifts the *uncertainty* score when the classifier is indecisive, is the normalised Shannon entropy $H(\mathbf{p})$ of the softmax vector $\mathbf{p}$:

$$H(\mathbf{p}) = -\frac{1}{\log K} \sum_{i=1}^{K} p_i \log p_i,$$

where p_i is the softmax probability assigned to class i and $K = 3$ is the number of sentiment classes (positive, negative, neutral); normalising by $\log K$ bounds $H(\mathbf{p})$ to $[0, 1]$. (ii) A *lightweight aspect extractor*, a keyword-driven approximation of aspect-based sentiment analysis (Pontiki et al., 2014), matches sentences against six financial aspect categories (revenue/earnings, growth/demand, risk/liquidity, valuation, product/AI, macro/policy) to attach evidence snippets to each record. (iii) A *sentiment-price correlation engine* computes Pearson and Spearman correlation, configurable lag analysis, Granger causality (Granger, 1969) with an augmented Dickey-Fuller stationarity pre-check (Dickey and Fuller, 1979), rolling correlation, and market adjustment, against same-day returns. We report evaluations of (i) and (iii) in Section 5 transparently, including their null and negative results.

5. Results

5.1. Stock FinBERT on the benchmark

Table 2 reports stock FinBERT on the 338-item benchmark. Accuracy is 53.0% and macro-F1 49.8%. The per-class pattern is the reverse of the neutral deficit reported for curated news text: FinBERT *over*-predicts neutral on this multi-source sample, assigning 228 items to the neutral class against 135 gold neutral items. Neutral recall is therefore high (81.5%) but neutral precision is low (48.2%), and the main errors are positive and negative items labelled neutral (74 of 122 positive and 44 of 81 negative items; Table 3). Positive recall is 28.7% and negative recall 42.0%. The keyword-lexicon baseline, run unchanged on the same 338 items, reaches 48.8% accuracy and 46.7% macro-F1. Its macro-F1 is 3.1 points below stock FinBERT, but an item-level paired bootstrap does not distinguish the two (95% CI $[-9.6, +3.4]$, $p = 0.36$), so we make no claim that FinBERT outperforms the baseline on this benchmark. As a stronger general-purpose baseline we also ran Twitter-RoBERTa (`cardiffnlp/twitter-roberta-base-sentiment-latest`), without fine-tuning and with the model card's preprocessing (user mentions and links replaced), on the same 338 items. It

reaches 56.8% macro-F1 and 59.2% accuracy, with positive, negative and neutral recall of 45.1%, 40.7% and 83.0% (Table 4). Its 7.0-point macro-F1 advantage over stock FinBERT is nominally significant (95% CI [+0.1, +14.0], unadjusted $p = 0.047$) but not after Holm correction ($p = 0.19$; Table 5).

Table 2. Stock FinBERT per-class performance on the 338-item benchmark (support: positive 122, negative 81, neutral 135; maximum length 128 tokens).

Class	Precision	Recall	F1	Predicted
Positive	72.9%	28.7%	41.2%	48
Negative	54.8%	42.0%	47.6%	62
Neutral	48.2%	81.5%	60.6%	228
Accuracy 53.0%; macro-F1 49.8%				

Table 3. Confusion matrix for stock FinBERT on the 338-item benchmark (rows: true label; columns: predicted).

	Pred. positive	Pred. negative	Pred. neutral
True positive	**35**	13	74
True negative	3	**34**	44
True neutral	10	15	**110**

5.2. Explainability as hypothesis generation

LIME is used here to illustrate, not to establish, a candidate mechanism. The positive and negative examples are drawn verbatim from the labelled evaluation set of the deployed system (`labeled_dataset.py`); the neutral example is a variant of a neutral sentence in that set. Explanations use stock FinBERT with 5,000 perturbation samples, whitespace tokens and a fixed random state, and show the eight highest-weighted tokens. Applied to unambiguous sentences, LIME assigns the highest weights to semantically appropriate tokens: "surged", "significantly" and "raised" for a positive prediction at 95.0% confidence (Figure 1), and "layoffs" and "loss" for a negative prediction at 96.9% confidence (Figure 2). For the neutral sentence *"The Federal Reserve kept interest rates unchanged, maintaining its cautious outlook on inflation."*, FinBERT predicts *negative* (55.9% confidence, against 27.5% for neutral), and LIME attributes this mainly to "cautious" and "inflation", followed by "outlook" and "unchanged", words that signal risk in isolation but are contextually neutral here (Figure 3). This is the misclassified neutral example requested by Reviewer 2, and it shows the failure mode directly: risk-laden vocabulary pulls a factual policy statement away from neutral. These examples suggested the hypothesis that exposure to neutral text containing sentiment-loaded vocabulary would help more than exposure to neutral text in general. They are single, hand-chosen explanations, and the hypothesis is tested directly by the comparison of arms (a) and (b) below.

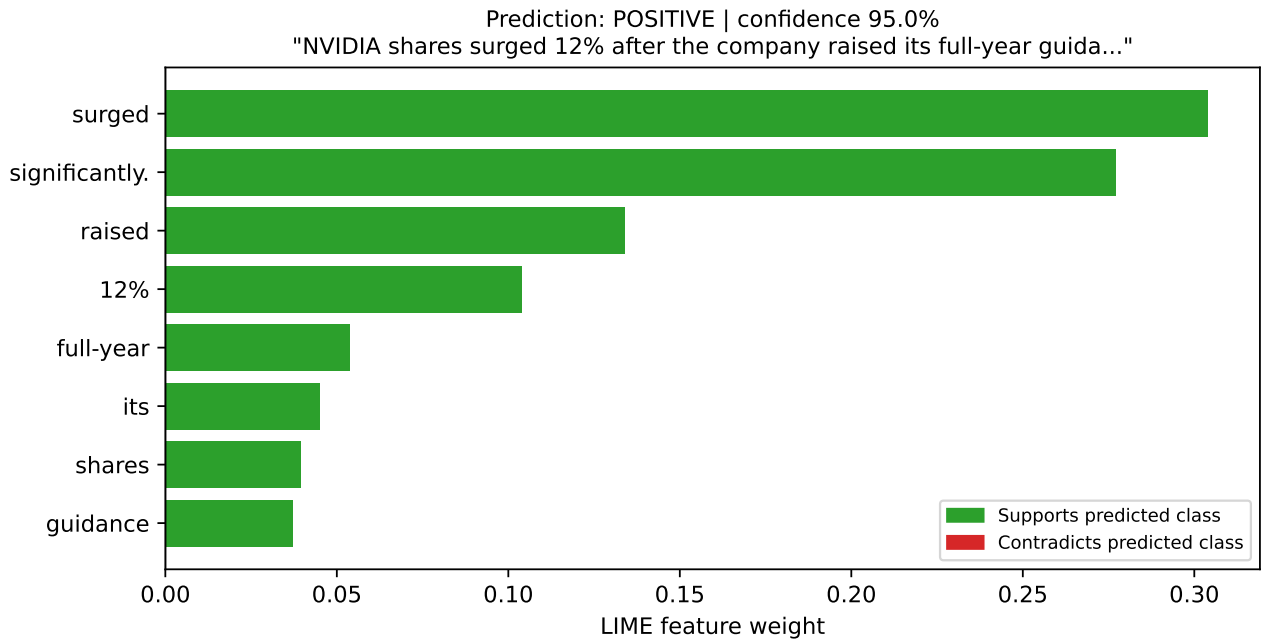


Figure 1. LIME token weights for a positive financial sentence (illustrative). Green bars indicate tokens supporting the *positive* prediction; red bars indicate tokens pushing against it. Stock FinBERT predicted *positive* at 95.0% confidence; *surged*, *significantly* and *raised* carry the largest weights.

5.3. Fine-tuning arms on the benchmark

Table 4 summarises the 15 fine-tuning runs. All three arms raise macro-F1 above stock FinBERT (49.8%): to 59.1% for the lexicon-loaded arm (a), 60.7% for the random-neutral arm (b) and 53.4% for the mixed arm (c), averaged over five seeds. The gains come from the polarised classes. Every arm raises positive and negative recall and lowers neutral recall from the stock 81.5%, to 47.7% (a), 60.6% (b) and 26.5% (c), while neutral precision rises from 48.2% to 63.1%, 64.1% and 76.8% respectively. The arms predict neutral for 102.6 (a), 128.0 (b) and 46.8 (c) items on average, against 135 gold neutral items and 228 for stock FinBERT. Arm (c) is also the least stable across seeds (macro-F1 standard deviation 3.7 points). The off-the-shelf Twitter-RoBERTa baseline (56.8%) does not beat the two best fine-tuned arms: it is 2.3 points below arm (a) and 3.9 points below arm (b), and 3.4 points above arm (c), but none of these differences is statistically distinguishable (95% CIs $[-8.8, +4.2]$, $[-10.1, +2.3]$ and $[-3.2, +10.1]$; Holm $p = 0.67$ for each; Table 5). Domain fine-tuning of FinBERT therefore matches, but is not shown to exceed, a general social-media sentiment model.

Table 5 gives the paired comparisons. Against stock FinBERT, the item-level bootstrap supports gains for arm (a) (+9.3 points of macro-F1, 95% CI $[+3.7, +15.1]$, Holm-adjusted $p = 0.004$) and arm (b) (+11.0 points, $[+5.6, +16.5]$, Holm-adjusted $p < 0.001$), but not for arm (c) (+3.6 points, $[-2.6, +10.0]$, Holm-adjusted $p = 0.51$). The central comparison for the LIME hypothesis, lexicon-loaded versus random neutrals, shows no advantage for lexical targeting: arm (a) is 1.6 points *below* arm (b) ($[-5.0, +1.7]$, Holm-adjusted $p = 0.51$; seed-paired t-test $p = 0.20$, Wilcoxon $p = 0.31$). Both neutral arms outperform the mixed arm in the bootstrap (Holm-adjusted $p = 0.012$ for (a) and 0.004

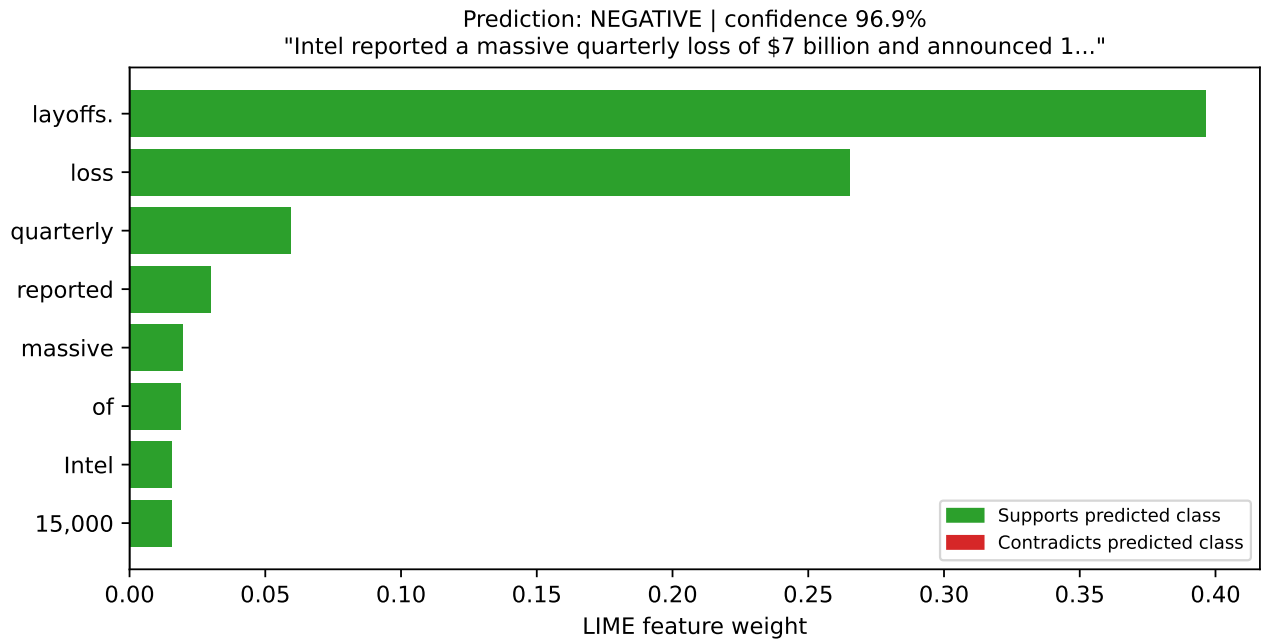


Figure 2. LIME token weights for a negative financial sentence (illustrative): *"Intel reported a massive quarterly loss of $7 billion and announced 15,000 layoffs."* Stock FinBERT predicted *negative* at 96.9% confidence; *layoffs* and *loss* are the principal drivers.

for (b)); the seed-paired tests point the same way but do not survive correction (Holm-adjusted t-test $p = 0.072$ for both; Wilcoxon $p = 0.0625$ before correction, the minimum attainable with five seeds). At the level of single runs, exact McNemar tests against stock reach uncorrected $p < 0.05$ in all five arm (b) runs and one arm (a) run, and in no arm (c) run; after Holm correction across the 15 runs only one (arm (b), seed 21; adjusted $p = 0.019$) remains below 0.05. Gains are thus clearer when seeds are pooled than for any individual run.

Table 6 breaks the benchmark down by channel. The channels differ in class mix (news is 57% neutral, Twitter 24%) and in ticker concentration (the five most frequent tickers cover 48% of news items but 24% of Twitter items). Stock FinBERT and the keyword baseline are strongest on news and weakest on Twitter, and the fine-tuning gains are concentrated on Twitter, where arms (a) and (b) reach 63.5% and 63.3% macro-F1 against 39.9% for stock; on news only arm (b) exceeds stock, and on Reddit no system exceeds 52%. Each channel has only 108–115 items, with 17 to 29 items in the smallest class, so these per-channel figures are descriptive and were not tested for significance. Most benchmark news items are short headlines that came from a public financial-news dataset ordered alphabetically by ticker, which the system's ingestion read only partially. As a result, news tickers are concentrated early in the alphabet (66% of news items have tickers beginning with 'A'; the top five tickers account for 47.8%), and news-channel results may not generalise to the wider range of companies covered in financial news. As a sensitivity check using the existing per-item predictions, excluding the 76 news items with 'A' tickers leaves 262 items, on which arms (a) and (b) still exceed stock FinBERT (mean macro-F1 57.3% and 56.8% against 44.1%; bootstrap gains of 13.2 and 12.7 points, 95% CIs $[+6.7, +20.0]$ and $[+6.2, +19.2]$), so the main conclusion is unchanged; the 39 remaining news items

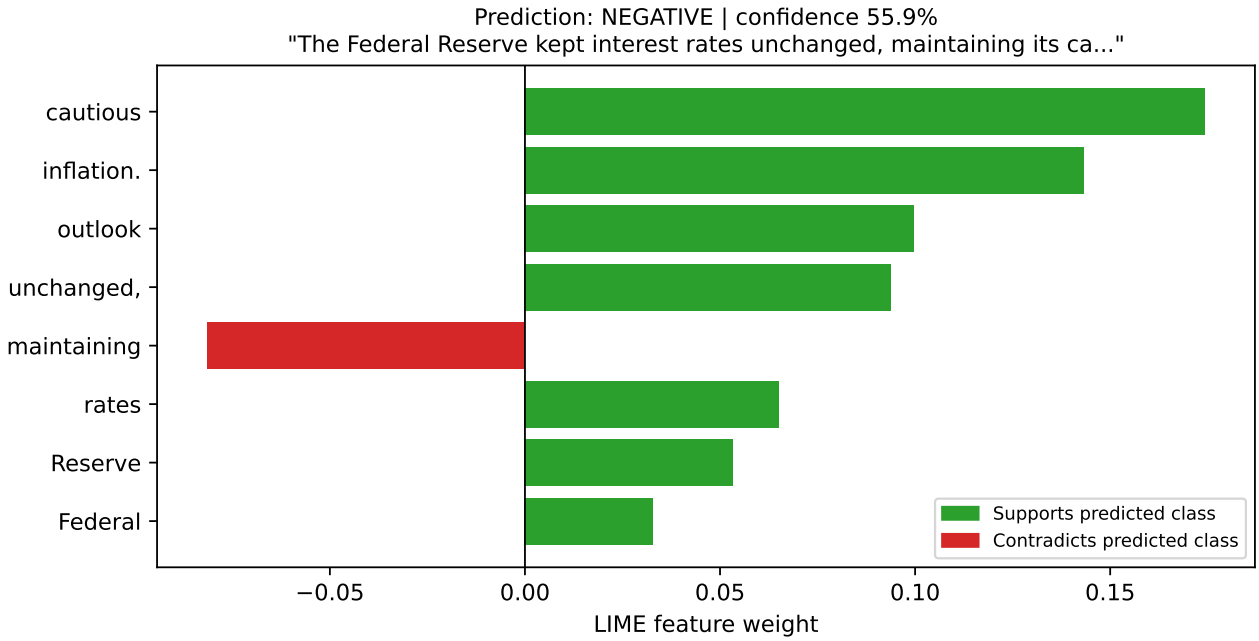


Figure 3. LIME token weights for a gold-neutral financial sentence that stock FinBERT misclassifies (illustrative). Stock FinBERT assigned 55.9% probability to *negative* against 27.5% to *neutral*; *cautious* and *inflation* are the top drivers towards the negative prediction.

are too few to separate any system from stock FinBERT.

Table 7 aggregates the errors that involve the neutral class. Stock FinBERT and Twitter-RoBERTa err mainly by absorbing polar items into neutral (118 and 102 such errors), and their few errors on gold-neutral items lean slightly negative (15 vs 10 and 14 vs 9). Fine-tuning reverses the direction: errors into neutral fall to 38.2 (a), 46.2 (b) and 11.0 (c), while gold-neutral items are pushed out of neutral, mostly towards positive for arms (a) and (c) (45.4 and 59.6 items) and about equally in both directions for arm (b) (27.0 negative, 26.2 positive).

5.4. Out-of-domain checks

Table 8 reports the out-of-domain evaluations. On both external datasets the fine-tuned models score below stock FinBERT. On the Twitter Financial News validation split, macro-F1 falls from 66.8% (stock) to 57.2% for arm (a) (−9.6 points, bootstrap 95% CI [−11.5, −7.7]), 63.0% for arm (b) (−3.8, [−5.5, −2.1]) and 39.7% for arm (c) (−27.1, [−29.2, −25.0]). On Financial PhraseBank, where stock FinBERT's scores are in-sample, the fine-tuned models also fall below stock on both subsets, with the same ordering ((b) above (a) above (c)). In each case the fall is concentrated in the neutral class: on Twitter Financial News, where 1,566 of 2,388 items are neutral, stock FinBERT predicts neutral for 1,388 items, against 747 (a), 1,019 (b) and 215 (c) for the fine-tuned models, and neutral recall falls from 75.6% to 44.3%, 59.4% and 13.2%.

Domain shift. Taken together, the in-domain and out-of-domain results indicate that fine-tuning adapts FinBERT to the class balance and labelling conventions of the target data rather than repair-

Table 4. Fine-tuned FinBERT on the 338-item benchmark: mean ± standard deviation over five seeds, with 95% t-intervals for macro-F1 and accuracy. Stock FinBERT shown for reference, with the off-the-shelf Twitter-RoBERTa baseline (single deterministic run).

Model	Macro-F1	Accuracy	Rec. pos.	Rec. neg.	Rec. neu.
Stock	49.8%	53.0%	28.7%	42.0%	81.5%
Twitter-RoBERTa	56.8%	59.2%	45.1%	40.7%	83.0%
(a) Lexicon-loaded	59.1 ± 0.9 [58.0, 60.3]	58.9 ± 0.9 [57.9, 60.0]	63.3 ± 2.3	71.1 ± 2.4	47.7 ± 4.0
(b) Random-neutral	60.7 ± 1.5 [58.9, 62.6]	60.8 ± 1.5 [59.0, 62.6]	53.1 ± 5.4	72.8 ± 1.5	60.6 ± 1.7
(c) Mixed	53.4 ± 3.7 [48.8, 58.0]	55.0 ± 2.8 [51.5, 58.5]	68.4 ± 7.2	82.2 ± 2.4	26.5 ± 9.0

Table 5. Item-level paired bootstrap of seed-averaged macro-F1 on the benchmark (B = 10,000; differences in percentage points; Holm correction over the six comparisons in the upper block; the Twitter-RoBERTa comparisons in the lower block form a separate family of four).

Comparison	Difference	95% CI	Holm p
(a) vs stock	+9.3	[+3.7, +15.1]	0.004
(b) vs stock	+11.0	[+5.6, +16.5]	< 0.001
(c) vs stock	+3.6	[−2.6, +10.0]	0.514
(a) vs (b)	−1.6	[−5.0, +1.7]	0.514
(a) vs (c)	+5.7	[+2.0, +9.6]	0.012
(b) vs (c)	+7.4	[+3.5, +11.4]	0.004
Twitter-RoBERTa vs stock	+7.0	[+0.1, +14.0]	0.190
Twitter-RoBERTa vs (a)	−2.3	[−8.8, +4.2]	0.672
Twitter-RoBERTa vs (b)	−3.9	[−10.1, +2.3]	0.672
Twitter-RoBERTa vs (c)	+3.4	[−3.2, +10.1]	0.672

ing a general weakness. On the project's multi-source benchmark, where stock FinBERT over-predicts neutral, the arms shift predictions toward the polarised classes and seed-averaged macro-F1 rises by 9 to 11 points for the two neutral arms. On Financial PhraseBank and Twitter Financial News, whose texts are predominantly professionally written or headline-style and whose neutral class is large, the same shift away from neutral lowers neutral recall and overall performance below stock FinBERT. The mixed arm (c), which adds polarised examples rather than neutrals, shifts furthest and transfers worst. Gains on an in-domain benchmark should therefore not be read as evidence of better financial sentiment classification in general; whether a given adaptation helps depends on how closely the deployment distribution and label definitions match the fine-tuning data.

5.5. Supporting analyses

Sentiment-price correlation. A demonstrative run of the correlation engine across a nine-ticker panel (90-day window, same-day alignment) produced modest, heterogeneous, and mostly non-significant effects. Same-day Pearson coefficients ranged from $r = -0.10$ (SPY) to +0.24 (NVDA);

Table 6. Benchmark composition and macro-F1 (%) by channel. Arms are means over five seeds. Tickers are the stored ticker tag; every item has one. Per-channel macro-F1 is descriptive only (small n, no significance tests).

	News	**Reddit**	**Twitter**	**All**
Items (pos / neg / neu)	32 / 17 / 66	38 / 29 / 41	52 / 35 / 28	122 / 81 / 135
Post dates	2017-08 – 2026-07	2021-02 – 2026-08	2022-01 – 2026-04	2017-08 – 2026-08
Distinct tickers	49	47	79	152
Top-5 ticker share	47.8%	38.0%	24.3%	20.1%
Stock FinBERT	67.7	42.4	39.9	49.8
(a) Lexicon-loaded	60.8	48.6	63.5	59.1
(b) Random-neutral	70.7	43.8	63.3	60.7
(c) Mixed	57.0	46.2	48.1	53.4
Twitter-RoBERTa	62.4	51.7	51.4	56.8
Keyword baseline	64.3	40.2	35.5	46.7

Table 7. Neutral-related error counts on the benchmark (135 gold-neutral, 122 positive, 81 negative items). Arms are means over five seeds.

	Gold neutral predicted as			**Into neutral**		
Model	**neu**	**neg**	**pos**	**pos→neu**	**neg→neu**	**pos↔neg**
Stock FinBERT	110	15	10	74	44	16
(a) Lexicon-loaded	64.4	25.2	45.4	24.0	14.2	30.0
(b) Random-neutral	81.8	27.0	26.2	32.2	14.0	33.0
(c) Mixed	35.8	39.6	59.6	8.0	3.0	42.0
Twitter-RoBERTa	112	14	9	61	41	13
Keyword baseline	91	25	19	64	34	31

NVDA, AAPL and AMZN were individually significant at $\alpha = 0.05$ (uncorrected), but under a Bonferroni correction for nine tickers ($\alpha_B = 0.0056$) none reached significance, and no Granger test attained $p < 0.05$ at one- or two-day lags. SPY's near-zero coefficient is the expected null for a diversified index tracker. These magnitudes are consistent with the literature that sentiment-return effects are statistically detectable but economically small (Tetlock, 2007; Kraaijeveld and De Smedt, 2020), and the rolling-correlation standard deviations (0.14 to 0.34) show the relationship is unstable across sub-windows. We report this as an honest null or modest result rather than selecting favourable windows.

Finance-emotion enrichment. Because no public benchmark annotates financial text at this emotion granularity, an accuracy evaluation is not possible; we instead ran a component-contribution ablation measuring how often removing a component flips the dominant emotion label on the frozen 338-item benchmark, with FinBERT class probabilities supplied to the taxonomy. Zeroing the entropy term changes 36.1% of labels (122/338), whereas removing the domain lexicon changes 2.1% (7/338). Under the full configuration the labels are dominated by "uncertainty" (250/338). An earlier version of

Table 8. Out-of-domain evaluation (macro-F1 / accuracy; neutral recall in parentheses). One seed per arm, chosen by validation macro-F1. Stock FinBERT was fine-tuned on Financial PhraseBank, so its PhraseBank scores are in-sample.

Model	PhraseBank all-agree (n = 2,264)	PhraseBank 50%-agree (n = 4,846)	Twitter Fin. News (val.) (n = 2,388)
Stock	96.3 / 97.2 (96.7)	88.3 / 89.0 (85.8)	66.8 / 72.5 (75.6)
(a) seed 84	92.0 / 93.5 (91.8)	82.0 / 82.2 (74.0)	57.2 / 57.8 (44.3)
(b) seed 84	93.3 / 94.9 (94.4)	84.4 / 85.5 (80.4)	63.0 / 65.8 (59.4)
(c) seed 21	77.9 / 76.8 (64.4)	67.1 / 64.6 (42.9)	39.7 / 38.6 (13.2)

this ablation did not pass the class probabilities to the taxonomy; this has been corrected. Because no emotion ground truth is available, these figures measure each component's influence, not its accuracy, and the enrichment layer is therefore reported as deployed but only partially validated.

System performance. FinBERT batch inference required 7.7 s for the 338 benchmark items (≈23 ms per text; median of five runs, batch size 32, 8-thread CPU, no GPU); the keyword baseline was effectively instantaneous (<5 ms for all 338 items); LIME explanations took 15 to 40 s, consistent with an on-demand rather than bulk feature.

6. Discussion

6.1. Interpretation and significance

Three findings stand out. First, on a benchmark sampled from the deployed system's own multi-source data, stock FinBERT does not show the neutral-recall deficit reported on curated news text; it over-predicts neutral (228 predictions against 135 gold items), and its main errors are polarised posts labelled neutral. The direction of FinBERT's neutral-class error therefore depends on the text distribution and the label definitions, and a single benchmark is not enough to characterise it.

Second, fine-tuning on a modest amount of in-domain labelled data improves in-domain performance: both neutral arms raise seed-averaged macro-F1 by roughly 9 to 11 points, and the gains survive an item-level paired bootstrap with Holm correction. The improvement comes from the polarised classes, at the cost of neutral recall.

Third, the LIME-motivated selection of lexicon-loaded neutrals gave no advantage over randomly chosen neutrals; if anything, the random-neutral arm scored slightly higher and transferred better out of domain. The LIME examples remain useful for illustrating how individual risk words can shift a prediction, but in this study they did not identify a data-selection strategy that outperformed a random control. We therefore present LIME as hypothesis-generating rather than as the source of the improvement. More broadly, the results illustrate the value of testing an explanation-derived hypothesis against a matched control, rather than reporting the intervention alone; this is the use of XAI as a diagnostic feedback mechanism that surveys identify as under-explored (Yeo et al., 2025), including when the feedback is negative.

The out-of-domain results qualify the in-domain gains. All fine-tuned models score below stock

FinBERT on Financial PhraseBank and on Twitter Financial News, with the largest drops for the mixed arm. Adaptation to one distribution's class balance and labelling conventions carries a cost elsewhere, so in-domain gains should be reported together with out-of-domain checks.

6.2. Threats to validity and limitations

We state limitations plainly. (i) *Label provenance.* Benchmark and training labels were produced by model-assisted labelling under a written rubric, with human audit by one author. Audit agreement was substantial (Landis and Koch, 1977) ($\kappa = 0.807$ on 31 held-out reserve items) and neutral precision in the training pool was 23/29 on a 30-item blind check, but these audits are small and the labels are not a multi-annotator consensus. Systematic labelling conventions are shared by the benchmark and the training arms, which may inflate in-domain gains relative to independently labelled data. (ii) *Benchmark composition.* The 338-item benchmark is mostly a channel-stratified random sample, but includes a negative-lexicon top-up (56 items) and is modest in size. (iii) *Statistical power.* Five seeds per arm give low power for seed-level tests (the exact Wilcoxon test cannot reach $p < 0.0625$), and seed-paired arm comparisons do not survive Holm correction; item-level tests pool seeds and should be read with that in mind. (iv) *Out-of-domain evidence.* The out-of-domain evaluations use a single seed per arm, chosen by validation macro-F1, so they do not capture seed variance; stock FinBERT's PhraseBank scores are in-sample. (v) *Arm design.* Arms (a) and (b) share a 600-item positive/negative base, a deviation from the planned neutral-only arms made after a pilot collapsed to all-neutral predictions; arm (c) consequently differs in class proportions, so its contrast with the other arms mixes class prior with data type. (vi) *Lost original data.* The fine-tuning corpus and run behind the earlier version of this study could not be recovered, and the original 200-item benchmark is not used; the earlier fine-tuning result is superseded, not replicated. (vii) *LIME stability.* LIME explanations are local linear approximations, can vary across runs and can in principle be adversarially fooled (Slack et al., 2020); the examples shown are illustrative. (viii) *Data imbalance.* The Twitter subcorpus is smaller and shallower than the Reddit and news subcorpora because of X API v2 quota constraints, so Twitter is a secondary signal and Twitter-only views carry lower statistical power. (ix) *Correlation is not causation.* The correlation engine is an exploratory tool; short-window sentiment series are noisy and non-stationary, macro events confound sentiment and price simultaneously, and the post-2008 liquidity regime weakens fundamental-to-price linkages. The reported null Granger results should be read accordingly. (x) *Enrichment validation.* The finance-emotion layer is deployed but lacks an accuracy evaluation against an annotated emotion corpus, which does not yet exist for these classes.

6.3. Ethical considerations

All data are publicly available and anonymised at ingestion, in line with platform terms of service and the institutional ethical framework. Two broader issues warrant note. First, a sentiment-analytics tool confers an informational advantage on its users, connecting to long-standing debates on market efficiency (Fama, 1970). Second, where natural-language summaries of sentiment are generated, model outputs can present statistical artefacts as confident narratives; outputs should therefore be accompanied by explicit documentation of intended use and limitations, an approach consistent with the model-cards framework (Mitchell et al., 2019). This study is for academic purposes; commercial deployment would require additional governance.

7. Conclusions

This paper set out to explain and reduce FinBERT's neutral-class error on multi-source investor text, and to test whether a LIME-derived hypothesis could guide the choice of fine-tuning data. On a 338-item benchmark drawn from the deployed system's database, stock FinBERT over-predicted neutral (neutral recall 81.5%, precision 48.2%) rather than under-predicting it. Fine-tuning on 900-item arms that share a positive/negative base raised seed-averaged macro-F1 from 49.8% to 59.1% (lexicon-loaded neutrals) and 60.7% (random neutrals), with gains supported by a paired item-level bootstrap after Holm correction; a mixed arm gave no reliable gain. Selecting neutrals by the sentiment-loaded vocabulary that LIME highlighted gave no advantage over a random-neutral control. On two out-of-domain datasets, every fine-tuned model scored below stock FinBERT, mainly through lower neutral recall. The evidence therefore supports in-domain adaptation with an out-of-domain cost, not a general repair, and it shows the value of testing explanation-derived hypotheses against matched controls and reporting paired significance tests.

Future work follows directly from the limitations. The most valuable next steps are (i) independent multi-annotator labelling of the benchmark and training data; (ii) multi-seed out-of-domain evaluation and adaptation methods that limit out-of-domain loss; (iii) extending the baseline comparison beyond the keyword lexicon and Twitter-RoBERTa to current financial LLMs (Wu et al., 2023; Fatouros et al., 2023); (iv) constructing an annotated financial-emotion corpus to evaluate the enrichment layer formally; and (v) feeding the resulting multi-source sentiment signal into hybrid temporal graph models for price prediction (Lynch et al., 2025).

Reproducibility. The analysis code is available from the authors on reasonable request (see the Data availability statement). It comprises the benchmark sampling scripts and the frozen 338-item benchmark with its labels; the build script and manifests for the three 900-item training arms; the fine-tuning runner and batch driver used for the 15 runs (three arms × five seeds, 13, 21, 42, 84 and 168), whose per-run outputs record the selected epoch, validation and benchmark metrics and per-item benchmark predictions and logits for every epoch; the statistics script (exact McNemar tests, item-level paired bootstrap with $B = 10{,}000$ and fixed seed 20261024, seed-paired tests and Holm correction) with its outputs; the Financial PhraseBank and Twitter Financial News evaluation scripts and outputs; the keyword-baseline run on the benchmark; and a locked list of CPU package versions. Model checkpoints are saved for the three retrained models used in the out-of-domain evaluations (arms (a) and (b), seed 84; arm (c), seed 21), which reproduced their original benchmark predictions exactly; checkpoints for the remaining runs were not retained. Also included are the Twitter-RoBERTa baseline (`results_v2/twitter_roberta_baseline/run_twitter_roberta.py`), the LIME rerun (`results_v2/lime_v2/run_lime_v2.py`), the per-source and neutral-error analyses (`results_v2/partial_items/run_partial_items.py`) and the enrichment ablation and timing rerun (`results_v2/enrichment_timing_v2/run_enrichment_timing_v2.py`).

Author contributions

D. Hubble: conceptualisation, methodology, software, data curation, formal analysis, investigation, visualisation, writing – original draft. S. Lynch: conceptualisation, supervision, validation, writing –

review and editing. Both authors have read and approved the final version of the manuscript.

Use of Generative-AI tools declaration

Generative AI models were used as part of the research method, to assign sentiment labels under the written rubric v2 (Section 3.2). The labels of the 338-item evaluation benchmark were produced by Anthropic Claude (claude-opus-5-5). The labels of the 2,300-item fine-tuning candidate pool (Section 3.3) were produced by OpenAI GPT-6 Astra (gpt-6-astra, high reasoning effort), except one batch of 50 items, which was produced by OpenAI gpt-6.1-sol (high reasoning effort). In both cases the models were not shown FinBERT outputs or any human labels. One author (D.H.) audited the labels. For the benchmark, he reviewed labels during rubric development and labelled a held-out reserve set, with Cohen's $\kappa = 0.807$ (87.1% agreement) on its 31 classifiable items; items he judged unjudgeable were removed. For the fine-tuning pool, he blind-relabelled 30 model-neutral items (23/30 agreement; neutral precision 0.793). No item-level label overrides were recorded, so all retained labels are the model labels. The authors also used AI assistants to help write analysis scripts and to edit the manuscript text; all code, outputs and text were checked by the authors. The authors designed the study, verified all results and conclusions, and take full responsibility for the content. No AI tool is listed as an author. Assistive-AI language tools (grammar and spelling checkers) are not generative tools and are not within the scope of this declaration.

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Ethics statement

The text analysed in this study was collected from publicly available sources (Reddit, X/Twitter and online financial news) and was anonymised at the point of ingestion, with author handles, display names and geolocation removed; no personally identifying information is reported. Under the Loughborough University ethical framework, this use of secondary, publicly available data did not require formal ethical review. The wider project, including its human-participant component, received a favourable ethical opinion from the Loughborough University Ethics Review Sub-Committee through the University's ethical review system, with one of the authors (S.L.) as the named responsible investigator.

Conflict of interest

All authors declare no conflicts of interest in this paper.

Data availability

The evaluation benchmark, derived results, and analysis code that support the findings of this study are available from the authors on reasonable request. The underlying social-media and news text is subject to the respective platforms' terms of service and cannot be redistributed in raw form.

References

Antweiler W and Frank MZ (2004) Is all that talk just noise? The information content of internet stock message boards. *The Journal of Finance* 59: 1259–1294. https://doi.org/10.1111/j.1540-6261.2004.00662.x

Araci D (2019) FinBERT: Financial sentiment analysis with pre-trained language models. arXiv:1908.10063. https://doi.org/10.48550/arXiv.1908.10063

Baker M and Wurgler J (2006) Investor sentiment and the cross-section of stock returns. *The Journal of Finance* 61: 1645–1680. https://doi.org/10.1111/j.1540-6261.2006.00885.x

Bollen J, Mao H and Zeng X (2011) Twitter mood predicts the stock market. *Journal of Computational Science* 2: 1–8. https://doi.org/10.1016/j.jocs.2010.12.007

Cohen J (1960) A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* 20: 37–46. https://doi.org/10.1177/001316446002000104

Da Z, Engelberg J and Gao P (2015) The sum of all FEARS investor sentiment and asset prices. *The Review of Financial Studies* 28: 1–32. https://doi.org/10.1093/rfs/hhu072

Deng X, Bashlovkina V, Han F, Baumgartner S and Bendersky M (2023) LLMs to the moon? Reddit market sentiment analysis with large language models. in *Companion Proceedings of the ACM Web Conference 2023* 1014–1019. https://doi.org/10.1145/3543873.3587605

Devlin J, Chang MW, Lee K and Toutanova K (2019) BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019* 4171–4186. https://doi.org/10.18653/v1/N19-1423

Dickey DA and Fuller WA (1979) Distribution of the estimators for autoregressive time series with a unit root.
Journal of the American Statistical Association 74: 427–431. https://doi.org/10.2307/2286348

Dietterich TG (1998) Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation* 10: 1895–1923. https://doi.org/10.1162/089976698300017197

Doshi-Velez F and Kim B (2017) Towards a rigorous science of interpretable machine learning. arXiv:1702.08608. https://doi.org/10.48550/arXiv.1702.08608

Efron B and Tibshirani RJ (1993) *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York, 1993. https://doi.org/10.1201/9780429246593

Fama EF (1970) Efficient capital markets: A review of theory and empirical work. *The Journal of Finance* 25: 383–417. https://doi.org/10.2307/2325486

Fatouros G, Soldatos J, Kouroumali K, Makridis G and Kyriazis D (2023) Transforming sentiment analysis in the financial domain with ChatGPT. *Machine Learning with Applications* 14 (2023), 100508. https://doi.org/10.1016/j.mlwa.2023.100508

Garcia D (2013) Sentiment during recessions. *The Journal of Finance* 68: 1267–1300. https://doi.org/10.1111/jofi.12027

Granger CWJ (1969) Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* 37: 424–438. https://doi.org/10.2307/1912791

Huang AH, Wang H and Yang Y (2023) FinBERT: A large language model for extracting information from financial text. *Contemporary Accounting Research* 40: 806–841. https://doi.org/10.1111/1911-3846.12832

Kraaijeveld O and De Smedt J (2020) The predictive power of public Twitter sentiment for forecasting cryptocurrency prices. *Journal of International Financial Markets, Institutions and Money* 65: 101188. https://doi.org/10.1016/j.intfin.2020.101188

Landis JR and Koch GG (1977) The measurement of observer agreement for categorical data. *Biometrics* 33: 159–174. https://doi.org/10.2307/2529310

Leippold M (2023) Sentiment spin: Attacking financial sentiment with GPT-3. *Finance Research Letters* 55: 103957. https://doi.org/10.1016/j.frl.2023.103957

Li D, Li Y, Chen X and Zhang W (2026) Augmenting large language models for financial sentiment analysis: A heuristic sparse mixture-of-experts framework. *PeerJ Computer Science* 12: e3607. https://doi.org/10.7717/peerj-cs.3607

Long S, Lucey B, Xie Y and Yarovaya L (2023) "I just like the stock": The role of Reddit sentiment in the GameStop share rally. *Financial Review* 58: 19–37. https://doi.org/10.1111/fire.12328

Lopez-Lira A and Tang Y (2023) Can ChatGPT forecast stock price movements? return predictability and large language models. SSRN Working Paper 4412788. https://doi.org/10.2139/ssrn.4412788

Loughran T and McDonald B (2011) When is a liability not a liability? textual analysis, dictionaries, and 10-Ks. *The Journal of Finance* 66: 35–65. https://doi.org/10.1111/j.1540-6261.2010.01625.x

Lundberg SM and Lee SI (2017) A unified approach to interpreting model predictions. in *Advances in Neural Information Processing Systems 30* 4765–4774. https://doi.org/10.48550/arXiv.1705.07874

Lynch ST, Derakhshan P and Lynch S (2025) A novel hybrid temporal fusion transformer graph neural network model for stock market prediction. *AppliedMath* 5: 176. https://doi.org/10.3390/appliedmath5040176

Malo P, Sinha A, Korhonen P, Wallenius J and Takala P (2014) Good debt or bad debt: Detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology* 65: 782–796. https://doi.org/10.1002/asi.23062

Mitchell M, Wu S, Zaldivar A, Barnes P, Vasserman L, Hutchinson B, Spitzer E, Raji ID and Gebru T (2019) Model cards for model reporting. *Proceedings of the Conference on Fairness, Accountability, and Transparency* 220–229. https://doi.org/10.1145/3287560.3287596

Pontiki M, Galanis D, Pavlopoulos J, Papageorgiou H, Androutsopoulos I and Manandhar S (2014) SemEval-2014 task 4: Aspect based sentiment analysis, in *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)* 27–35. https://doi.org/10.3115/v1/S14-2004

Ribeiro MT, Singh S and Guestrin C (2016) "Why should I trust you?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 1135–1144. https://doi.org/10.1145/2939672.2939778

Rudin C (2019) Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence* 1, 206–215. https://doi.org/10.1038/s42256-019-0048-x

Schumaker RP and Chen H (2009) Textual analysis of stock market prediction using breaking financial news: The AZFinText system. *ACM Transactions on Information Systems* 27: 1–19. https://doi.org/10.1145/1462198.1462204

Slack D, Hilgard S, Jia E, Singh S and Lakkaraju H (2020) Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* 180–186. https://doi.org/10.1145/3375627.3375830

Sravankumar B, Krishna G and Rao M (2025) Integration of explainable AI with deep learning for real-time sentiment-driven stock price prediction. *Computational Economics*. https://doi.org/10.1007/s10614-025-11159-w

Tetlock PC (2007) Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance* 62:1139–1168. https://doi.org/10.1111/j.1540-6261.2007.01232.x

Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L and Polosukhin I (2017) Attention is all you need. in *Advances in Neural Information Processing Systems 30:* 6000–6010. https://doi.org/10.48550/arXiv.1706.03762

Vismara S, Latifi G, Meinzinger L and Pass A (2026) Generative AI-powered venture screening: Can large language models help venture capitalists? *International Review of Financial Analysis* 109: 104748. https://doi.org/10.1016/j.irfa.2025.104748

Wu S, Irsoy O, Lu S, Dabravolski V, Dredze M, Gehrmann S, Kambadur P, Rosenberg D and Mann G (2023) BloombergGPT: A large language model for finance. arXiv:2303.17564. https://doi.org/10.48550/arXiv.2303.17564

Yang H, Uy MCS and Huang A (2020) FinBERT: A pretrained language model for financial communications. arXiv:2006.08097. https://doi.org/10.48550/arXiv.2006.08097

Yang H, Chen H, Jiang C, He J and Li P (2026) PLTA-FinBERT: Pseudo-label generation-based test-time adaptation for financial sentiment analysis. *Big Data and Cognitive Computing* 10: 59. https://doi.org/10.3390/bdcc10020059

Yeo WJ, van der Heever W, Mao R, Cambria E, Satapathy R and Mengaldo G (2025) A comprehensive review on financial explainable AI. *Artificial Intelligence Review* 58. https://doi.org/10.1007/s10462-024-11077-7

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